Deriving difficulty reweightings

CS336 Assignment 5, Section 5.3

`derive_difficulty_reweightings`

Source and scope. These notes follow the Spring 2026 handout, pp. 28–29, especially its equations (39)–(43). Throughout, rewards are binary, the loss normalizer is the constant $Z = G$, and we take $G \to \infty$. Thus part (b) concerns GRPO with *constant* loss normalization.

Write $\eta_\theta(x) = \mathbb{E}_{y\sim\pi_\theta(\cdot|x)}[r(y \mid x)]$. The answers are the following prompt weights, evaluated at the current policy and held fixed when taking a gradient:

Part	Method	$w(x, \text{stopgrad}(\pi_\theta))$
(a)	Dr. GRPO	1
(b)	GRPO, constant loss normalization	$\dfrac{1}{\sqrt{\eta_\theta(x)(1-\eta_\theta(x))}}$
(c)	MaxRL	$\dfrac{1}{\eta_\theta(x)}$

For (b), assume $0 < \eta_\theta(x) < 1$; for (c), assume $\eta_\theta(x) > 0$. The last page discusses zero denominators and the finite-group case.

Notation. Fix a prompt x and the current parameters θ_0. Sample $Y_1, \ldots, Y_G$ independently from $\pi_{\theta_0}(\cdot \mid x)$, and set

$$R_j = r(Y_j \mid x), \qquad S_j = \nabla_\theta \log \pi_\theta(Y_j \mid x)|_{\theta=\theta_0}, \qquad \mu_G = \frac{1}{G}\sum_{j=1}^{G} R_j. \tag{1}$$

Here S_j is a vector, and $\eta = \eta_{\theta_0}(x)$ is the probability of a correct response. Higher η means an easier prompt for this policy. Assume that the prompt distribution and reward function do not depend on θ, and that differentiation and expectation can be interchanged.

Two identities used in every part. The score has conditional mean zero:

$$\begin{aligned}\mathbb{E}[S_j \mid x] &= \sum_y \pi_\theta(y \mid x)\nabla_\theta \log \pi_\theta(y \mid x)\Big|_{\theta_0} \\ &= \nabla_\theta \sum_y \pi_\theta(y \mid x)\Big|_{\theta_0} = \nabla_\theta 1 = 0.\end{aligned} \tag{2}$$

Multiplying by reward gives the policy-gradient identity:

$$\mathbb{E}[R_j S_j \mid x] = \sum_y r(y \mid x)\nabla_\theta \pi_\theta(y \mid x)\Big|_{\theta_0} = \nabla_\theta \eta_\theta(x)|_{\theta_0}. \tag{3}$$

Consequently, subtracting the population mean does not change this expectation:

$$\mathbb{E}[(R_j - \eta)S_j \mid x] = \mathbb{E}[R_j S_j \mid x] - \eta\mathbb{E}[S_j \mid x] = \nabla_\theta \eta_\theta(x)|_{\theta_0}. \tag{4}$$

One calculation for all three normalizers. Let d_G denote the group normalizer: 1, the group standard deviation, or μ_G. At this fixed prompt, the estimator is

$$\begin{aligned}\widehat{g}_G(x) &= \frac{1}{G}\sum_{j=1}^{G}\frac{R_j-\mu_G}{d_G}S_j \\ &= \frac{1}{d_G}\left(\frac{1}{G}\sum_{j=1}^{G}R_jS_j - \mu_G\frac{1}{G}\sum_{j=1}^{G}S_j\right).\end{aligned} \tag{5}$$

Suppose $d_G \xrightarrow{\text{a.s.}} d(x) > 0$. The law of large numbers, applied to each vector coordinate, gives

$$\mu_G \xrightarrow{\text{a.s.}} \eta, \qquad \frac{1}{G}\sum_j S_j \xrightarrow{\text{a.s.}} 0, \qquad \frac{1}{G}\sum_j R_jS_j \xrightarrow{\text{a.s.}} \nabla_\theta\eta_\theta(x)|_{\theta_0}. \tag{6}$$

Substituting these limits into (5),

$$\boxed{\widehat{g}_G(x) \xrightarrow{\text{a.s.}} \frac{1}{d(x)}\,\nabla_\theta\eta_\theta(x)|_{\theta_0}.} \tag{7}$$

The required moments must exist; averaging this limit over prompts also requires the corresponding integrability/interchange conditions.

To match the handout's stop-gradient surrogate, freeze the reference policy:

$$J_w(\theta;\theta_0) = \mathbb{E}_{x\sim\rho}[w(x,\pi_{\theta_0})\eta_\theta(x)]. \tag{8}$$

Its gradient at the current policy is

$$\nabla_\theta J_w(\theta;\theta_0)|_{\theta_0} = \mathbb{E}_{x\sim\rho}\Big[w(x,\pi_{\theta_0})\,\nabla_\theta\eta_\theta(x)|_{\theta_0}\Big]. \tag{9}$$

Comparing (7) and (9) gives $w = 1/d$.

(a) Dr. GRPO. Here $d_G = 1$, so $d(x) = 1$ and

$$\boxed{w_{\text{Dr. GRPO}}(x,\pi_{\theta_0}) = 1.} \tag{10}$$

The mean baseline cancels in expectation by (4). In the infinite-group limit, the estimator is therefore the gradient of ordinary expected reward, with the original prompt distribution ρ.

(b) GRPO with constant loss normalization. Since $R_j \in \{0,1\}$, $R_j^2 = R_j$. Hence

$$\mathrm{Var}(R_j \mid x) = \mathbb{E}[R_j^2 \mid x] - \mathbb{E}[R_j \mid x]^2 = \eta - \eta^2 = \eta(1-\eta). \tag{11}$$

The group standard deviation converges to $\sqrt{\eta(1-\eta)}$. This is true whether the sample variance uses denominator G or $G-1$. Thus

$$\boxed{w_{\text{GRPO}}(x,\pi_{\theta_0}) = \frac{1}{\sqrt{\eta_{\theta_0}(x)(1-\eta_{\theta_0}(x))}}.} \tag{12}$$

(c) MaxRL. Here $d_G = \mu_G \xrightarrow{\text{a.s.}} \eta$. Therefore

$$\boxed{w_{\text{MaxRL}}(x,\pi_{\theta_0}) = \frac{1}{\eta_{\theta_0}(x)}.} \tag{13}$$

Equivalently, its population gradient is $\mathbb{E}_{x\sim\rho}[\nabla_\theta \log \eta_\theta(x)]$ at $\theta = \theta_0$, where the log and expectation are well-defined.

Why `stopgrad` matters. In (8), θ is differentiated and θ_0 is held fixed. This is precisely the meaning of the handout's $w(x, \text{stopgrad}(\pi_\theta))$. The derivative does not include an extra $\eta_\theta(x)\nabla_\theta w$ term.

MaxRL makes the distinction especially clear:

$$J_{\text{MaxRL}}(\theta;\theta_0) = \mathbb{E}_x\left[\frac{\eta_\theta(x)}{\eta_{\theta_0}(x)}\right], \tag{14}$$

$$J_{\text{MaxRL}}(\theta_0;\theta_0) = 1, \tag{15}$$

$$\nabla_\theta J_{\text{MaxRL}}(\theta;\theta_0)|_{\theta_0} = \mathbb{E}_x\left[\frac{\nabla_\theta \eta_\theta(x)|_{\theta_0}}{\eta_{\theta_0}(x)}\right]. \tag{16}$$

Although the surrogate's value at the current parameters is 1, its derivative with respect to the first argument need not be zero. Simplifying the expression to $\eta_\theta/\eta_\theta = 1$ before differentiating would incorrectly let the denominator vary. For a scalar illustration, $f(p;p_0) = p/p_0$ satisfies $f(p_0;p_0) = 1$, but $\partial f/\partial p = 1/p_0$.

Connection to the finite-group baseline bias. At finite G, μ_G contains R_j, so it is not independent of S_j. Using independence only for the other samples,

$$\begin{aligned}\mathbb{E}[\mu_G S_j \mid x] &= \frac{1}{G}\mathbb{E}[R_j S_j \mid x] + \frac{1}{G}\sum_{k\neq j}\mathbb{E}[R_k \mid x]\mathbb{E}[S_j \mid x] \\ &= \frac{1}{G}\,\nabla_\theta \eta_\theta(x)|_{\theta_0}\,. \end{aligned} \tag{17}$$

Therefore the finite-group Dr. GRPO estimator has expectation

$$\mathbb{E}[\widehat{g}_G^{\text{Dr. GRPO}}(x) \mid x] = \left(1 - \frac{1}{G}\right)\nabla_\theta \eta_\theta(x)|_{\theta_0}\,. \tag{18}$$

The factor tends to 1, explaining why it does not appear in part (a). For (b) and (c), the denominator is also random and correlated with the numerator at finite G. Replacing an expectation of a ratio by a ratio of expectations is unjustified; the preceding derivation uses the large-group limit instead.

Reading the weights. Dr. GRPO adds no difficulty-dependent multiplier. GRPO's multiplier is smallest at $\eta = 1/2$ and grows toward both endpoints. MaxRL's multiplier decreases as η grows, emphasizing harder prompts relative to easier ones. For example:

Success probability η	Dr. GRPO	GRPO	MaxRL
0.1	1	3.33	10
0.5	1	2	2
0.9	1	3.33	1.11

These numbers multiply each prompt's own gradient $\nabla_\theta \eta_\theta(x)$. A larger weight alone does not imply a larger resulting gradient norm.

Zero denominators and implementation. Without stabilization, GRPO is undefined at $\eta = 0$ or 1, and MaxRL is undefined at $\eta = 0$. Even when $0 < \eta < 1$, a finite group can contain identical rewards; all centered advantages then vanish. Adding $\varepsilon > 0$ to the normalizer makes those groups contribute zero. For fixed ε, the population weights become

$$w_{\text{GRPO},\varepsilon}(x) = \frac{1}{\sqrt{\eta(x)(1-\eta(x))} + \varepsilon}, \qquad w_{\text{MaxRL},\varepsilon}(x) = \frac{1}{\eta(x) + \varepsilon}. \tag{19}$$

The boxed answers use the problem's unstabilized formulas. They also rely on constant loss normalization: dividing each response by its own length introduces a response-dependent factor, which this derivation does not remove.